Cluster Analysis

Distance & Similarity Measures

A beginner-friendly teaching handout with one worked example from start to finish

What you will learn

- How to measure *how far apart* or *how alike* two observations are.
- The main distance measures (Euclidean, Manhattan, Minkowski, Chebyshev) and similarity measures (cosine, Jaccard, simple matching).
- How to compute a distance matrix and a similarity matrix by hand, using one small dataset.
- Why these matrices are the starting point of hierarchical clustering.

1 The Big Idea: Distance vs. Similarity

Cluster analysis groups observations so that items *inside* a group are alike and items in *different* groups are unlike. Before any grouping can happen we need a number that tells us how alike (or unlike) two observations are.

Key definitions

- **Distance (dissimilarity)** $d(x, y)$: *small* when observations are close, *large* when they are far apart.
- **Similarity** $s(x, y)$: *large* when observations are alike, *small* when they are different.

Each observation is described by p variables: $x = (x_1, x_2, \ldots, x_p)$ and $y = (y_1, y_2, \ldots, y_p)$.

They are two sides of the same coin, and one can be converted into the other:

$$d = 1 - s \qquad \text{or} \qquad s = \frac{1}{1+d} \qquad \text{or} \qquad s = 1 - \frac{d}{d_{\max}}$$

1.1 What makes a "proper" distance (a metric)?

Property	Meaning
Non-negativity	$d(x, y) \geq 0$
Identity	$d(x, y) = 0$ exactly when $x = y$
Symmetry	$d(x, y) = d(y, x)$
Triangle inequality	$d(x, z) \leq d(x, y) + d(y, z)$

2 Main Distance Measures

2.1 Euclidean distance — "as the crow flies"

The straight-line distance: the Pythagorean theorem extended to p dimensions.

$$d_E(x, y) = \sqrt{\sum_{k=1}^{p} (x_k - y_k)^2}$$

Use when: variables are numeric and on comparable scales. It is the default in most clustering software.

2.2 Manhattan (city-block) distance — "walking along streets"

$$d_M(x, y) = \sum_{k=1}^{p} |x_k - y_k|$$

Use when: you want less sensitivity to one large difference, or the variables behave like separate "blocks" (counts, grid-like movement).

2.3 Minkowski distance — the general family

$$d_r(x, y) = \left(\sum_{k=1}^{p} |x_k - y_k|^r \right)^{1/r}, \qquad r \geq 1$$

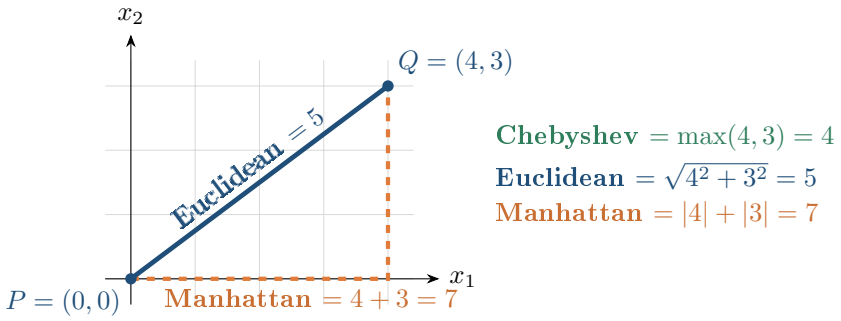
Here p is the number of variables and r is the *order*. Special cases:

- $r = 1$ gives **Manhattan**; $r = 2$ gives **Euclidean**; $r \to \infty$ gives **Chebyshev**.

2.4 Chebyshev distance — "the biggest single gap"

$$d_\infty(x, y) = \max_k |x_k - y_k|$$

Use when: the worst single difference is what matters (e.g. quality-control tolerances).



Three ways to measure the distance between the same two points P and Q.

2.5 Other basic measures

Measure	Formula	Notes
Squared Euclidean	$\sum_k (x_k - y_k)^2$	No square root; used by Ward's method
Hamming	number of positions with $x_k \neq y_k$	Binary / categorical data; counts mismatches
Canberra	$\sum_k \frac{\lvert x_k - y_k \rvert}{\lvert x_k \rvert + \lvert y_k \rvert}$	Sensitive to values near zero
Mahalanobis	$\sqrt{(x-y)^\top S^{-1}(x-y)}$	Accounts for correlations and scales (S = covariance matrix)

Scale warning

If one variable is measured in thousands and another in tenths, the large one dominates the distance. **Standardize** the variables (e.g. z-scores) first when units differ.

3 Main Similarity Measures

3.1 Cosine similarity — compares *direction*, ignores *size*

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \, \|y\|} = \frac{\sum_k x_k y_k}{\sqrt{\sum_k x_k^2} \sqrt{\sum_k y_k^2}}$$

Range: -1 to 1 (0 to 1 when all values are non-negative). The related **cosine distance** is $1 - \cos(x, y)$.

Use when: the *pattern* matters more than the amount: text mining, recommender systems, preference profiles. For example $(1, 2, 3)$ and $(2, 4, 6)$ have cosine similarity exactly 1, even though their Euclidean distance is not 0.

3.2 Binary data: the 2×2 table

For two binary vectors, count matches and mismatches over the p variables:

	$y = 1$	$y = 0$
$x = 1$	a	b
$x = 0$	c	d

with $a + b + c + d = p$.

3.3 Simple Matching Coefficient (SMC)

$$\mathrm{SMC}(x, y) = \frac{a + d}{a + b + c + d}$$

Use when: both 1 and 0 are equally informative (**symmetric binary**), e.g. gender, or yes/no questions of equal importance.

3.4 Jaccard coefficient

$$J(x, y) = \frac{a}{a + b + c}$$

Use when: a shared 0 tells you nothing (**asymmetric binary**), e.g. "both own a rare item", "both have the symptom", shopping baskets. Joint absences (d) are ignored.

3.5 Others to know

Measure	Formula	Use
Sørensen–Dice	$\frac{2a}{2a + b + c}$	Like Jaccard but gives more weight to matches
Pearson correlation	r between the two profiles	Compares the *shape* of profiles; distance version $1 - r$
Gower similarity	average of per-variable similarities	**Mixed** data (numeric and categorical)

Quick guide: which measure should I use?

Your data	Good choice
Numeric, same units, magnitude matters	Euclidean (or Manhattan)
Numeric, different units	Standardize, then Euclidean
Numeric, only the pattern matters	Cosine or correlation
Binary, 0 and 1 equally meaningful	Simple matching
Binary, presence is what matters	Jaccard
Mixed types	Gower

4 Our Teaching Dataset

Six students record the **weekly hours** they spend on three activities. We use this *same* dataset for every calculation below.

Student	X_1 Reading	X_2 Sport	X_3 Music
A	1	2	3
B	2	4	5
C	6	4	1
D	7	6	3
E	3	1	6
F	4	3	8

All variables are in the same unit (hours), so no standardization is needed. Each student is a point in 3-D space, e.g. $A = (1, 2, 3)$ and $B = (2, 4, 5)$.

5 Euclidean Distance: Step by Step

Recipe (with $p = 3$ variables)

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}$$

(1) subtract variable by variable **(2)** square each difference **(3)** add them up **(4)** take the square root.

Example 1: A and B

Data: $A = (1, 2, 3)$, $B = (2, 4, 5)$

1. Formula

$$d(A, B) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

2. Substitution

$$d(A, B) = \sqrt{(1 - 2)^2 + (2 - 4)^2 + (3 - 5)^2}$$

3. Calculation

$$d(A, B) = \sqrt{(-1)^2 + (-2)^2 + (-2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9}$$

4. Final answer

$$d(A, B) = 3.000$$

5. Interpretation

A and B are close: B simply spends a bit more time on every activity.

Example 2: B and C

Data: $B = (2, 4, 5)$, $C = (6, 4, 1)$

1. Formula

$$d(B, C) = \sqrt{(b_1 - c_1)^2 + (b_2 - c_2)^2 + (b_3 - c_3)^2}$$

2. Substitution

$$d(B, C) = \sqrt{(2 - 6)^2 + (4 - 4)^2 + (5 - 1)^2}$$

3. Calculation

$$d(B, C) = \sqrt{(-4)^2 + 0^2 + 4^2} = \sqrt{16 + 0 + 16} = \sqrt{32}$$

4. Final answer

$$d(B, C) = 4\sqrt{2} \approx 5.657$$

5. Interpretation

Same sport hours, but very different reading and music habits (B loves music, C loves reading). Much farther apart than A and B.

Example 3: C and D

Data: $C = (6, 4, 1)$, $D = (7, 6, 3)$

1. Formula

$$d(C, D) = \sqrt{(c_1 - d_1)^2 + (c_2 - d_2)^2 + (c_3 - d_3)^2}$$

2. Substitution

$$d(C, D) = \sqrt{(6 - 7)^2 + (4 - 6)^2 + (1 - 3)^2}$$

3. Calculation

$$d(C, D) = \sqrt{(-1)^2 + (-2)^2 + (-2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9}$$

4. Final answer

$$d(C, D) = 3.000$$

5. Interpretation

C and D are both heavy readers with similar habits: a close pair, just like A and B.

Example 4: D and E

Data: $D = (7, 6, 3)$, $E = (3, 1, 6)$

1. Formula

$$d(D, E) = \sqrt{(d_1 - e_1)^2 + (d_2 - e_2)^2 + (d_3 - e_3)^2}$$

2. Substitution

$$d(D, E) = \sqrt{(7 - 3)^2 + (6 - 1)^2 + (3 - 6)^2}$$

3. Calculation

$$d(D, E) = \sqrt{4^2 + 5^2 + (-3)^2} = \sqrt{16 + 25 + 9} = \sqrt{50}$$

4. Final answer

$$d(D, E) = 5\sqrt{2} \approx 7.071$$

5. Interpretation

D reads and plays sport a lot; E does neither much. They are very different.

Example 5: E and F

Data: $E = (3, 1, 6)$, $F = (4, 3, 8)$

1. Formula

$$d(E, F) = \sqrt{(e_1 - f_1)^2 + (e_2 - f_2)^2 + (e_3 - f_3)^2}$$

2. Substitution

$$d(E, F) = \sqrt{(3 - 4)^2 + (1 - 3)^2 + (6 - 8)^2}$$

3. Calculation

$$d(E,F) = \sqrt{(-1)^2 + (-2)^2 + (-2)^2} = \sqrt{1+4+4} = \sqrt{9}$$

4. Final answer

$$d(E,F) = 3.000$$

5. Interpretation

Both are music-focused and low on sport: another close pair.

5.1 All 15 pairs at a glance

Pair	Differences $(x-y)$	Sum of squares	Distance
A–B	$(-1,-2,-2)$	9	**3.000**
A–C	$(-5,-2,2)$	33	**5.745**
A–D	$(-6,-4,0)$	52	**7.211**
A–E	$(-2,1,-3)$	14	**3.742**
A–F	$(-3,-1,-5)$	35	**5.916**
B–C	$(-4,0,4)$	32	**5.657**
B–D	$(-5,-2,2)$	33	**5.745**
B–E	$(-1,3,-1)$	11	**3.317**
B–F	$(-2,1,-3)$	14	**3.742**
C–D	$(-1,-2,-2)$	9	**3.000**
C–E	$(3,3,-5)$	43	**6.557**
C–F	$(2,1,-7)$	54	**7.348**
D–E	$(4,5,-3)$	50	**7.071**
D–F	$(3,3,-5)$	43	**6.557**
E–F	$(-1,-2,-2)$	9	**3.000**

6 The Euclidean Distance (Dissimilarity) Matrix

Place every pairwise distance in a square table (rows and columns in the order A, B, C, D, E, F):

$$D = \begin{pmatrix} 0 & 3.000 & 5.745 & 7.211 & 3.742 & 5.916 \\ 3.000 & 0 & 5.657 & 5.745 & 3.317 & 3.742 \\ 5.745 & 5.657 & 0 & 3.000 & 6.557 & 7.348 \\ 7.211 & 5.745 & 3.000 & 0 & 7.071 & 6.557 \\ 3.742 & 3.317 & 6.557 & 7.071 & 0 & 3.000 \\ 5.916 & 3.742 & 7.348 & 6.557 & 3.000 & 0 \end{pmatrix}$$

	A	B	C	D	E	F
A	0	3.000	5.745	7.211	3.742	5.916
B	3.000	0	5.657	5.745	3.317	3.742
C	5.745	5.657	0	3.000	6.557	7.348
D	7.211	5.745	3.000	0	7.071	6.557
E	3.742	3.317	6.557	7.071	0	3.000
F	5.916	3.742	7.348	6.557	3.000	0

Grey: diagonal (always 0). Orange: the closest pairs (A–B, C–D, E–F).

How to read the matrix

- The **diagonal is 0**: every student is at distance 0 from themselves.
- The matrix is **symmetric**: $d(A, B) = d(B, A)$. Only one triangle is needed: $\frac{n(n-1)}{2} = \frac{6 \cdot 5}{2} = 15$ distinct values.
- The **smallest** off-diagonal values (three pairs at 3.000) are the closest pairs; the **largest** (7.348, pair C–F) is the most distant pair.

7 Similarity Measures on the Same Data

7.1 Cosine similarity (numeric data)

$$\cos(x, y) = \frac{x_1 y_1 + x_2 y_2 + x_3 y_3}{\sqrt{x_1^2 + x_2^2 + x_3^2}\,\sqrt{y_1^2 + y_2^2 + y_3^2}}$$

Vector lengths:

$$\|A\| = \sqrt{14} \approx 3.742, \quad \|B\| = \sqrt{45} \approx 6.708, \quad \|C\| = \sqrt{53} \approx 7.280$$

$$\|D\| = \sqrt{94} \approx 9.695, \quad \|E\| = \sqrt{46} \approx 6.782, \quad \|F\| = \sqrt{89} \approx 9.434$$

Cosine example: A and B

$$A \cdot B = (1)(2) + (2)(4) + (3)(5) = 2 + 8 + 15 = 25$$

$$\cos(A, B) = \frac{25}{\sqrt{14}\,\sqrt{45}} = \frac{25}{25.100} \approx 0.996$$

Interpretation: almost identical *profiles* (both lean toward music, then sport, then reading).

Cosine example: C and E

$$C \cdot E = (6)(3) + (4)(1) + (1)(6) = 18 + 4 + 6 = 28$$

$$\cos(C, E) = \frac{28}{\sqrt{53}\sqrt{46}} = \frac{28}{49.376} \approx 0.567$$

Interpretation: quite different profiles (C favours reading, E favours music).

Pair	$x \cdot y$	$\|x\|\|y\|$	cos
A–B	25	25.100	0.9960
A–C	17	27.240	0.6241
A–D	28	36.277	0.7718
A–E	23	25.377	0.9063
A–F	34	35.299	0.9632
B–C	33	48.837	0.6757
B–D	53	65.038	0.8149
B–E	40	45.497	0.8792
B–F	60	63.285	0.9481

Pair	$x \cdot y$	$\|x\|\|y\|$	cos
C–D	69	70.583	0.9776
C–E	28	49.376	0.5671
C–F	44	68.680	0.6407
D–E	45	65.757	0.6843
D–F	70	91.466	0.7653
E–F	63	63.984	0.9846

7.2 The cosine similarity matrix (final similarity matrix)

$$S_{\cos} = \begin{pmatrix} 1 & 0.996 & 0.624 & 0.772 & 0.906 & 0.963 \\ 0.996 & 1 & 0.676 & 0.815 & 0.879 & 0.948 \\ 0.624 & 0.676 & 1 & 0.978 & 0.567 & 0.641 \\ 0.772 & 0.815 & 0.978 & 1 & 0.684 & 0.765 \\ 0.906 & 0.879 & 0.567 & 0.684 & 1 & 0.985 \\ 0.963 & 0.948 & 0.641 & 0.765 & 0.985 & 1 \end{pmatrix}$$

	A	B	C	D	E	F
A	1	0.996	0.624	0.772	0.906	0.963
B	0.996	1	0.676	0.815	0.879	0.948
C	0.624	0.676	1	0.978	0.567	0.641
D	0.772	0.815	0.978	1	0.684	0.765
E	0.906	0.879	0.567	0.684	1	0.985
F	0.963	0.948	0.641	0.765	0.985	1

Grey: diagonal (always 1). Orange: off-diagonal values ≥ 0.95.

7.3 Binary version for Jaccard and Simple Matching

These two measures need **0/1 data**. We define 1 = "regularly does this activity" (**at least 3 hours per week**), 0 otherwise.

Student	Reading	Sport	Music	Binary vector
A	0	0	1	$(0,0,1)$
B	0	1	1	$(0,1,1)$
C	1	1	0	$(1,1,0)$
D	1	1	1	$(1,1,1)$
E	1	0	1	$(1,0,1)$
F	1	1	1	$(1,1,1)$

Information loss

D and F become *identical* after converting to 0/1, even though their original hours differ. Turning numeric data into binary data **throws information away**.

Binary example: A = (0, 0, 1) and B = (0, 1, 1)

	$B=1$	$B=0$
$A=1$	$a=1$ (Music)	$b=0$
$A=0$	$c=1$ (Sport)	$d=1$ (Reading)

$$J(A,B) = \frac{a}{a+b+c} = \frac{1}{1+0+1} = \frac{1}{2} = 0.500$$

$$\text{SMC}(A,B) = \frac{a+d}{a+b+c+d} = \frac{1+1}{3} = \frac{2}{3} \approx 0.667$$

Interpretation: SMC is higher because it also rewards the shared "does not read regularly" ($d=1$); Jaccard ignores that joint absence.

Binary example: C = (1, 1, 0) and D = (1, 1, 1)

Here $a=2$, $b=0$, $c=1$, $d=0$:

$$J(C,D) = \frac{2}{2+0+1} = 0.667, \qquad \text{SMC}(C,D) = \frac{2+0}{3} = 0.667$$

They agree because there are no joint absences ($d=0$).

7.4 Jaccard similarity matrix

	A	B	C	D	E	F
A	1	0.500	0	0.333	0.500	0.333
B	0.500	1	0.333	0.667	0.333	0.667
C	0	0.333	1	0.667	0.333	0.667
D	0.333	0.667	0.667	1	0.667	1
E	0.500	0.333	0.333	0.667	1	0.667
F	0.333	0.667	0.667	1	0.667	1

7.5 Jaccard in detail: all 15 pairs

For each pair, a = activities both do $(1,1)$, b = only the first does $(1,0)$, c = only the second does $(0,1)$. Shared zeros (d) are ignored.

$$J(x,y) = \frac{a}{a+b+c}$$

Worked example: A $= (0,0,1)$ and B $= (0,1,1)$

$a = 1$ (both do Music), $b = 0$, $c = 1$ (only B does Sport), and $d = 1$ (neither reads, ignored).

$$J(A,B) = \frac{1}{1+0+1} = \frac{1}{2} = 0.500$$

Pair	a	b	c	$a+b+c$	J
A–B	1	0	1	2	0.500
A–C	0	1	2	3	0.000
A–D	1	0	2	3	0.333
A–E	1	0	1	2	0.500
A–F	1	0	2	3	0.333
B–C	1	1	1	3	0.333
B–D	2	0	1	3	0.667
B–E	1	1	1	3	0.333
B–F	2	0	1	3	0.667

Pair	a	b	c	$a+b+c$	J
C–D	2	0	1	3	0.667
C–E	1	1	1	3	0.333
C–F	2	0	1	3	0.667
D–E	2	1	0	3	0.667
D–F	3	0	0	3	1.000
E–F	2	0	1	3	0.667

7.6 Jaccard similarity matrix (final)

$$S_J = \begin{pmatrix} 1 & 0.500 & 0 & 0.333 & 0.500 & 0.333 \\ 0.500 & 1 & 0.333 & 0.667 & 0.333 & 0.667 \\ 0 & 0.333 & 1 & 0.667 & 0.333 & 0.667 \\ 0.333 & 0.667 & 0.667 & 1 & 0.667 & 1 \\ 0.500 & 0.333 & 0.333 & 0.667 & 1 & 0.667 \\ 0.333 & 0.667 & 0.667 & 1 & 0.667 & 1 \end{pmatrix}$$

- **A and C have similarity 0**: they share no activity at all.
- **D and F have similarity 1**: identical after binarizing, although their original hours differ (information loss).

7.7 Jaccard distance matrix

Convert to a distance with $d_J = 1 - J$ (the form hierarchical clustering needs), e.g. $d_J(A, B) = 0.500$ and $d_J(D, F) = 0$:

$$D_J = \begin{pmatrix} 0 & 0.500 & 1 & 0.667 & 0.500 & 0.667 \\ 0.500 & 0 & 0.667 & 0.333 & 0.667 & 0.333 \\ 1 & 0.667 & 0 & 0.333 & 0.667 & 0.333 \\ 0.667 & 0.333 & 0.333 & 0 & 0.333 & 0 \\ 0.500 & 0.667 & 0.667 & 0.333 & 0 & 0.333 \\ 0.667 & 0.333 & 0.333 & 0 & 0.333 & 0 \end{pmatrix}$$

7.8 Simple Matching similarity matrix

	A	B	C	D	E	F
A	1	0.667	0	0.333	0.667	0.333
B	0.667	1	0.333	0.667	0.333	0.667
C	0	0.333	1	0.667	0.333	0.667
D	0.333	0.667	0.667	1	0.667	1
E	0.667	0.333	0.333	0.667	1	0.667
F	0.333	0.667	0.667	1	0.667	1

SMC differs from Jaccard only where a pair shares a joint 0 (pairs A–B and A–E).

7.9 Converting distance into similarity

Using $s = \dfrac{1}{1+d}$ on the Euclidean results:

$$s(A, B) = \frac{1}{1+3} = 0.250, \qquad s(C, F) = \frac{1}{1+7.348} \approx 0.120$$

Identical points would give $s = 1$.

8 Distance Matrix vs. Similarity Matrix

Feature	Distance / dissimilarity matrix	Similarity matrix
Meaning of a value	How **far apart**	How **alike**
Diagonal	**0** (zero distance to self)	**1** (perfect similarity to self)
Closest pair	**Smallest** off-diagonal value	**Largest** off-diagonal value
Typical range	0 to ∞ (Euclidean)	0 to 1 (cosine: -1 to 1)
Symmetric?	Yes	Yes
Example here	$d(A, B) = 3.000$	$\cos(A, B) = 0.996$

Rule of thumb

For distance, small = close. For similarity, large = close. Always check which one your software expects. Most clustering routines need a *distance* matrix, so a similarity matrix is first converted, e.g. $d = 1 - s$.

Do the measures agree on our data? Euclidean says the closest pairs are A–B, C–D and E–F (all 3.000), and cosine also ranks these pairs very high (0.996, 0.978, 0.985). But they can disagree: cosine finds A and F quite alike (0.963) although their Euclidean distance is large (5.916), because cosine ignores *how much* time is spent and looks only at the *shape* of the profile.

9 From Distance Matrix to Hierarchical Clustering

Hierarchical (agglomerative) clustering **starts from the distance matrix** and repeats:

1. Start with every observation as its own cluster (6 clusters).
2. Find the **two closest clusters** in the matrix and merge them.
3. Update the distances between the new cluster and all the others.
4. Repeat until one cluster remains. The merge heights form the **dendrogram**.

Step 3 needs a rule for the distance between *clusters*, called the **linkage method**. For clusters P and Q:

$$d_{\text{single}}(P,Q) = \min_{x\in P,\, y\in Q} d(x,y), \qquad d_{\text{complete}}(P,Q) = \max_{x\in P,\, y\in Q} d(x,y)$$

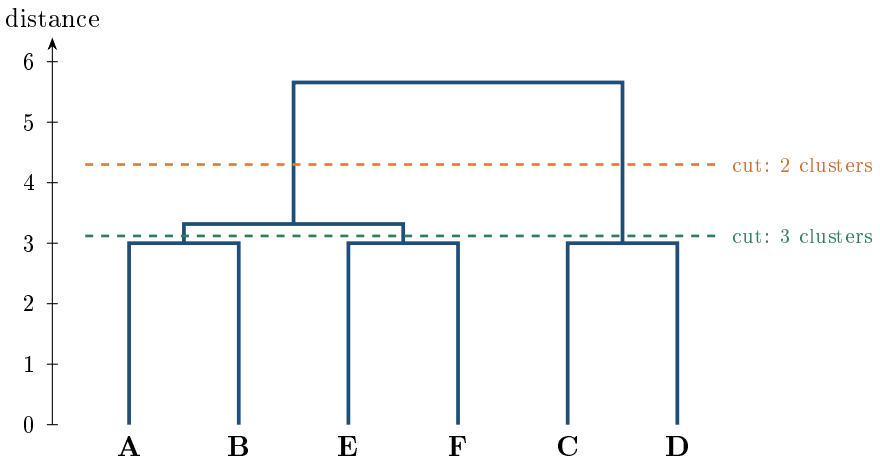
$$d_{\text{average}}(P,Q) = \frac{1}{|P|\,|Q|} \sum_{x\in P} \sum_{y\in Q} d(x,y)$$

Linkage	Rule	Tendency
Single	distance between the *nearest* pair	Long, "chained" clusters
Complete	distance between the *farthest* pair	Compact, similar-sized clusters
Average	mean of all pairwise distances	A compromise
Ward	merge that least increases the within-cluster sum of squares	Compact, equal-sized clusters

On our dataset the three smallest entries are all 3.000 (A–B, C–D, E–F), so these three pairs merge first. The distances between $\{A,B\}$ and $\{E,F\}$ are $d(A,E) = 3.742$, $d(A,F) = 5.916$, $d(B,E) = 3.317$, $d(B,F) = 3.742$, therefore

$$d_{\text{single}} = \min = 3.317, \qquad d_{\text{complete}} = \max = 5.916, \qquad d_{\text{average}} = \frac{3.742 + 5.916 + 3.317 + 3.742}{4} = 4.179$$

Linkage	Merge A–B, C–D, E–F	$\{A,B\} + \{E,F\}$	Final: $+\{C,D\}$
Single	3.000	3.317	5.657
Complete	3.000	5.916	7.348
Average	3.000	4.179	6.486



Single-linkage dendrogram of our six students. Dashed lines show where cutting the tree gives 2 or 3 clusters.

All three linkage methods give the same tree shape: $\{A, B, E, F\}$ join first and $\{C, D\}$ join last. A two-cluster cut gives $\{A, B, E, F\}$ and $\{C, D\}$; a three-cluster cut gives $\{A, B\}$, $\{E, F\}$ and $\{C, D\}$. Only the merge *heights* differ.

10 Complete Python Code

This script reproduces every number in this handout: the distance measures, the Euclidean matrix, the cosine, Jaccard and simple-matching matrices, and hierarchical clustering with dendrograms for three linkage methods. (Requires `numpy`, `pandas`, `matplotlib`, `scipy`.)

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.spatial.distance import squareform
from scipy.cluster.hierarchy import linkage, dendrogram

# ------------------------------------------------------------------
# 1. Data: weekly hours on Reading, Sport, Music
# ------------------------------------------------------------------
labels = ["A", "B", "C", "D", "E", "F"]
X = np.array([
    [1, 2, 3],   # A
    [2, 4, 5],   # B
    [6, 4, 1],   # C
    [7, 6, 3],   # D
    [3, 1, 6],   # E
    [4, 3, 8],   # F
], dtype=float)

# ------------------------------------------------------------------
# 2. Distance measures (written from scratch)
# ------------------------------------------------------------------
def euclidean(x, y):
    return np.sqrt(np.sum((x - y) ** 2))

def manhattan(x, y):
    return np.sum(np.abs(x - y))

def minkowski(x, y, r):
    return np.sum(np.abs(x - y) ** r) ** (1.0 / r)

def chebyshev(x, y):
    return np.max(np.abs(x - y))

# ------------------------------------------------------------------
# 3. Similarity measures (written from scratch)
```

```
# ---------------------------------------------------------------
def cosine_similarity(x, y):
    return np.dot(x, y) / (np.linalg.norm(x) * np.linalg.norm(y))

def jaccard(x, y):
    a = np.sum((x == 1) & (y == 1))          # joint presences
    denom = np.sum((x == 1) | (y == 1))      # a + b + c
    return a / denom if denom > 0 else np.nan

def simple_matching(x, y):
    return np.mean(x == y)                    # (a + d) / p

# ---------------------------------------------------------------
# 4. Build a full pairwise matrix from any pair function
# ---------------------------------------------------------------
def pairwise_matrix(data, func):
    n = data.shape[0]
    M = np.zeros((n, n))
    for i in range(n):
        for j in range(n):
            M[i, j] = func(data[i], data[j])
    return M

def show(M, title, decimals=3):
    print(f"\n{title}")
    print(pd.DataFrame(M, index=labels, columns=labels).round(decimals))

# ---------------------------------------------------------------
# 5. Step-by-step Euclidean examples (A-B, B-C, C-D, D-E, E-F)
# ---------------------------------------------------------------
name_to_row = {name: X[i] for i, name in enumerate(labels)}
for u, v in [("A", "B"), ("B", "C"), ("C", "D"), ("D", "E"), ("E", "F")]:
    diff = name_to_row[u] - name_to_row[v]
    ss = np.sum(diff ** 2)
    print(f"d({u},{v}): differences={diff}, squares={diff**2}, "
          f"sum={ss:.0f}, distance={np.sqrt(ss):.3f}")

# Other distance measures for A and B
A, B = X[0], X[1]
print("\nOther distances for A and B:")
print("Manhattan        :", manhattan(A, B))                 # 5
print("Minkowski (r=3)  :", round(minkowski(A, B, 3), 3))    # 2.571
print("Chebyshev        :", chebyshev(A, B))                 # 2

# ---------------------------------------------------------------
# 6. Euclidean distance matrix
# ---------------------------------------------------------------
D_euc = pairwise_matrix(X, euclidean)
show(D_euc, "Euclidean distance matrix")

# ---------------------------------------------------------------
# 7. Cosine similarity matrix (numeric data)
# ---------------------------------------------------------------
S_cos = pairwise_matrix(X, cosine_similarity)
show(S_cos, "Cosine similarity matrix")

# ---------------------------------------------------------------
# 8. Binary version (1 = at least 3 hours) for Jaccard and SMC
# ---------------------------------------------------------------
Xb = (X >= 3).astype(int)
print("\nBinary data (1 = at least 3 hours/week):")
print(pd.DataFrame(Xb, index=labels, columns=["Reading", "Sport", "Music"]))

S_jac = pairwise_matrix(Xb, jaccard)
S_smc = pairwise_matrix(Xb, simple_matching)
show(S_jac, "Jaccard similarity matrix")
D_jac = 1 - S_jac                        # Jaccard distance
show(D_jac, "Jaccard distance matrix (1 - Jaccard)")
show(S_smc, "Simple matching similarity matrix")

# ---------------------------------------------------------------
# 9. Convert similarity to distance (d = 1 - s) and distance to
#    similarity (s = 1 / (1 + d))
# ---------------------------------------------------------------
```

```
D_cos = np.clip(1 - S_cos, 0, None)   # clip tiny negative rounding errors
S_from_euc = 1 / (1 + D_euc)
show(D_cos, "Cosine distance matrix (1 - cosine)")
show(S_from_euc, "Similarity from Euclidean: 1 / (1 + d)")

# ---------------------------------------------------------------
# 10. Hierarchical clustering from the distance matrix
#     squareform() converts the square matrix into the condensed
#     form that scipy's linkage() expects.
# ---------------------------------------------------------------
condensed = squareform(D_euc, checks=False)

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
for ax, method in zip(axes, ["single", "complete", "average"]):
    Z = linkage(condensed, method=method)
    print(f"\nLinkage matrix ({method}): "
          "[cluster1, cluster2, merge height, size]")
    print(np.round(Z, 3))
    dendrogram(Z, labels=labels, ax=ax)
    ax.set_title(f"{method.capitalize()} linkage")
    ax.set_ylabel("Euclidean distance")
plt.tight_layout()
plt.show()
```

11 Key Takeaways

Remember

- **Distance** is small for similar observations; **similarity** is large for similar observations. Convert with $d = 1 - s$ or $s = 1/(1 + d)$.
- **Euclidean**, **Manhattan** and **Minkowski** work on numeric data; **standardize** first if units differ.
- **Cosine** compares direction (profile), **Jaccard** ignores joint absences, **simple matching** counts them.
- A **distance matrix** has 0 on the diagonal; a **similarity matrix** has 1.
- Hierarchical clustering uses the distance matrix plus a **linkage rule** (single, complete, average, Ward) to build the dendrogram.

11.1 Practice questions

1. Compute $d(A, E)$ by hand and check it against the matrix ($\sqrt{14} \approx 3.742$).
2. Compute the Manhattan distance between B and C and compare it with the Euclidean distance 5.657.
3. Why do D and F have Jaccard similarity 1 even though their hours differ? What does this tell you about binarizing data?
4. Which pair is most similar under cosine? Which pair is farthest under Euclidean? Do they agree?